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DIRECTORATE OF ONLINE EDUCATION

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR - 603 203

MCA III SEMESTER

MINI PROJECT VIVA VOCE

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ENROLMENT NUMBER : **EA2532251010500**

NAME OF STUDENT : **MOHAMED SHAHITH AKEEL S A**

PROJECT OVERVIEW

Title of the Mini Project

COMMUNITY ITEM LENDING SYSTEM

(CILS)

A simple web application to borrow and lend household tools, appliances, and equipment among neighbors

What is this Project?

Simple Goal: Helps neighbors borrow and lend tools and equipment directly within residential communities.

Less Waste: Cuts down clutter and saves money instead of buying items used only once or twice a year.

How Does it Work?

Online Website: Residents post items with photos; neighbors search, view availability, and click request.

No Duplicate Booking: When an owner approves a request, the system locks the item instantly as 'Lent'.

Who Gets the Benefit?

Community Borrowers: Families save money by borrowing power drills, lawnmowers, ladders, and camping gear.

Item Owners: Turn idle household items into community goodwill while maintaining full control over lending.

ABSTRACT

What the Project Solves and Delivers

1. The Problem Today

Idle Household Items: Power tools, pressure washers, and ladders sit unused 95% of the year in garages.

Cost Barrier: Buying expensive equipment for a one-off DIY repair or home project is wasteful.

Informal Chaos: Asking on WhatsApp groups leads to lost messages, forgotten returns, and damaged trust.

2. Our Solution

One Single Website: Verified residents list surplus items with photos, specs, and category tags.

Easy Requests: Borrowers click 'Request Item', pick borrow/return dates, and submit in one step.

Live Tracking: Automated state machine updates item status from Available to Lent to Returned.

3. Results Delivered

Zero Confusion: Real-time locking prevents double-booking so two neighbors never get promised the same tool.

Audit History: Full permanent log of all borrow requests, approval timestamps, and return dates.

Fast Process: Eliminates awkward back-and-forth messaging, enabling instant borrowing coordination.

Main Result: A complete, working web application built with Python Flask and TiDB MySQL, tested and ready to use.

INTRODUCTION

Why This Problem Matters — Current Situation

Background: What Happens in Households

Underutilized Tools: Average electric drill is used for only 12 to 15 minutes in its entire lifespan, sitting idle for years.

Unnecessary Expenses: Neighbors independently purchase identical lawnmowers, pressure washers, and ladders at high cost.

Garage Clutter: Valuable home storage space is crowded with rarely used DIY tools, sports gear, and camping equipment.

Environmental Waste: Manufacturing single-owner household appliances creates avoidable carbon emissions, scrap metal, and e-waste.

What Communities Need & Our Project Goals

Collaborative Sharing: Communities need a safe, hyper-local platform to share household tools freely without financial friction.

No Direct Platform: Without a dedicated system, residents hesitate to ask neighbors or fear unreturned/damaged equipment.

Project Goal 1: Build an intuitive, friendly website where verified neighbors post items and view community listings.

Project Goal 2: Make the lending process automated, fair, and transparent with real-time status and return records.

CAPABILITIES

Key Features of the System

1. User Authentication & Safe Profiles

Secure Accounts: Role-based access with encrypted passwords (PBKDF2:SHA256) and session protection.

Community Safety: Contact information (phone/email) shared only after mutual borrowing approval.

2. Clean Online Catalog with Photos

Easy Search: Instant keyword search and dynamic category filter buttons (Tools, Books, Sports, Electronics, Household).

Clear Availability: Real-time visual status badges (Available, Lent, Pending) on all item cards.

3. Automatic Lending & Booking Engine

No Overbooking: Real-time database state locking prevents overlapping borrow requests for the same item.

Safe Database: Atomic SQL transactions ensure data consistency across multiple simultaneous requests.

4. Return Tracking & Audit Records

One-Click Returns: Owner confirms return with one click, restoring item status to Available immediately.

Permanent History: Complete transaction ledger of past lending cycles, return timestamps, and borrower history.

SYSTEM REQUIREMENTS

Hardware and Software Requirements

Hardware Requirements

Processor: Intel Core i3 / i5 or AMD Ryzen (any standard PC or laptop)

RAM (Memory): 4 GB minimum (8 GB recommended for best development speed)

Hard Disk Space: 10 GB free space for database storage, photos, and project environment

Monitor / Display: Standard 15" or 14" laptop/desktop screen (1366x768 / 1080p resolution)

Keyboard & Mouse: Any standard computer keyboard and mouse / trackpad

Internet Connection: Basic broadband or mobile hotspot connection (2 Mbps+)

Printer: Normal office printer (optional for printing borrowing receipts/records)

Software Requirements

Operating System: Windows 10 / 11 or Linux (Ubuntu)

Programming Language: Python (Version 3.10 or higher)

Web Framework: Flask 3.0 (Lightweight and fast web engine)

Database Server: TiDB Cloud Serverless (MySQL 8.0 Protocol compliant)

Database Connector: PyMySQL (Safe Python connector for MySQL queries)

Security Engine: Werkzeug Security (PBKDF2 Password Hashing)

Frontend Stack: HTML5, CSS3, JavaScript, Bootstrap 5.3

Web Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox

Code Editor: Visual Studio Code (VS Code)

CURRENT APPROACH

Existing System and Its Problems

Step 1: Manual Requests

How it is done: Residents post requests in crowded WhatsApp groups, make phone calls, or knock on neighbor doors.

Problem: Messages get buried quickly, forgotten on busy days, or missed by neighbors who have the item.

Step 2: Excel / Memory Tracking

How it is done: Lenders try to remember who borrowed their tools or note them down on scraps of paper / Excel.

Problem: Return dates are forgotten; items remain unreturned for months or are lost permanently.

Step 3: Handover Confusion

How it is done: Items change hands without verifying current condition or recording borrow details.

Problem: No permanent record or audit proof exists for item return dates, damage accountability, or disputes.

Main Gaps in the Existing System

Double Booking: Because chats and spreadsheets are not synchronized, two neighbors ask for and are promised the same item simultaneously.

Too Slow: It takes days of back-and-forth messaging and awkward reminders just to coordinate borrowing a simple tool.

No Audit Proof: Without central logs, lenders cannot verify past borrower history, leading to mistrust and reluctance to share.

PROPOSED APPROACH

How the Proposed System Improves the Process

1. Easy Web Frontend

Simple Interface: Built using HTML5 and Bootstrap 5.3 so the website looks clean, modern, and uncluttered.

Works Anywhere: Runs smoothly on laptops, tablets, and smartphone web browsers.

Clear Alerts: Shows green pop-up banners when requests succeed and red warnings if an item is already reserved.

No Training Needed: Simple buttons like 'Browse Items' and 'Request Borrow' that any resident can use immediately.

2. Python Backend Logic

Fast Python Engine: Uses Python 3.10+ and Flask 3.0 to process requests and routes in milliseconds.

Safe Passwords: Passwords are encrypted with PBKDF2:SHA256 so accounts stay completely secure.

Photo Resizer: Automatically resizes and validates item photos into clean pictures for the catalog.

Auto Status Engine: Manages item states (Available -> Pending -> Lent -> Returned) automatically.

3. Secure TiDB Cloud MySQL

Organized Tables: Stores users, categories, items, and borrow requests cleanly with relational foreign keys.

Instant Stock Lock: Automatically locks item status upon approval, making duplicate booking impossible.

Reliable Storage: Data is securely hosted in serverless cloud MySQL, safe from server restarts or power loss.

Permanent History: Keeps a full audit log of every single borrowing transaction for transparent accountability.

SYSTEM DESIGN

System Design — Data Flow Diagram (DFD)

Figure 1: DFD Level 0 — Context Diagram

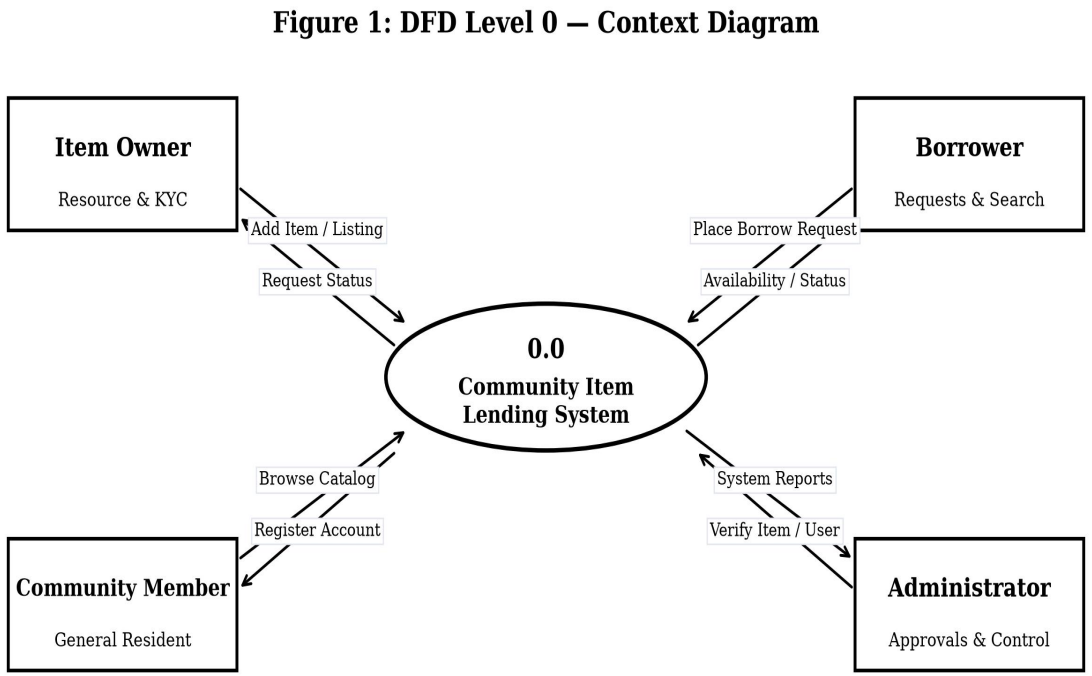


Figure: Data Flow Diagram (DFD) — How Data Moves in the System

How Data Moves in 4 Simple Steps

1. Item Owner: Enters household item details (tools, ladder, category) and uploads photo into the system.

2. System Database: Saves item records in TiDB MySQL and displays them on the live community catalog.

3. Community Borrower: Browses catalog, chooses needed item, and submits borrowing request with return date.

4. Review & Handover: Owner reviews request, clicks approve, system locks item as 'Lent', and shares contact details.

SYSTEM INTERACTIONS

Use Case Diagram — What Each User Can Do

Figure 4: UML Use Case Diagram

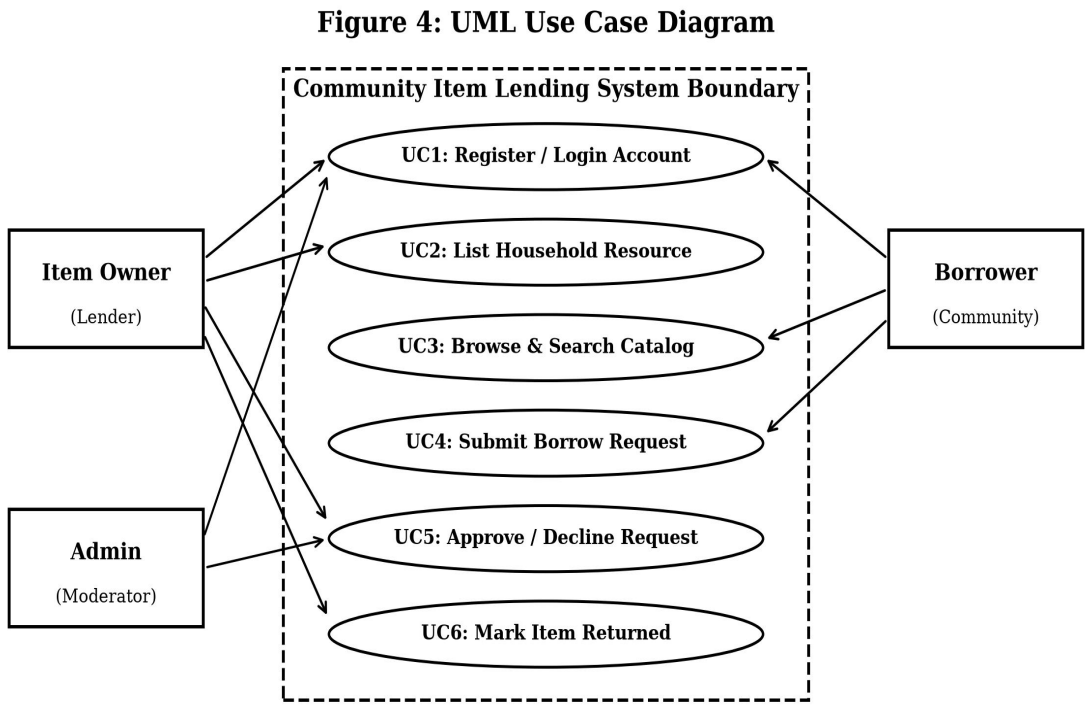


Figure: Use Case Diagram — Users and Their Actions

User Roles and Their Actions

Community Borrower:

- Register resident account and log in securely.
- Search and filter available items by category or keyword.
- Submit borrowing request with return date and track approval.

Item Owner:

- Log in to personal dashboard and inventory management.
- Add household items with photo, condition, and category.
- Approve or reject borrow requests and confirm returns.

MODULE DESCRIPTION

Modules at a Glance and Work Done

1. User Login & Registration

Handles account signup and secure login for residents using PBKDF2:SHA256 password hashing.

2. Public Item Catalog

Shows all available household items online with clear photos, categories, and availability badges.

3. Add Item Listing & Photo

Allows residents to list extra household items, choose category, write description, and upload photos.

4. Borrow Request Module

Lets neighbors submit borrowing requests with preferred borrow/return dates and purpose.

5. Stock Reservation Engine

Automatically locks approved items so they change status to Lent and cannot be double-booked.

6. Return Tracking & Audit Log

Enables owners to mark items as returned, restoring availability and logging transaction history.

SAMPLE SOURCE CODE

Sample Source Code — Approving a Request & Locking Stock

File: app/routes/requests.py (Python / Flask)

```python
@req_bp.route('/request/<int:id>/approve', methods=['POST'])
@login_required
def approve_request(id):
    # 1. Find the borrowing request
    req = BorrowRequest.query.get(id)
    item = Item.query.get(req.item_id)

    # 2. Check if item is available
    if item.status == 'Available':
        # 3. Lock item status safely
        item.status = 'Lent'
        req.status = 'Approved'

        # 4. Save changes to database
        db.session.commit()
        flash('Request approved! Item locked.', 'success')
    else:
        flash('Item is already lent out.', 'danger')

    return redirect('/dashboard')
```

What this code does:

Step 1: Finds the neighbor's borrow request and the requested household item.

Step 2: Checks if the item is currently available in the community database.

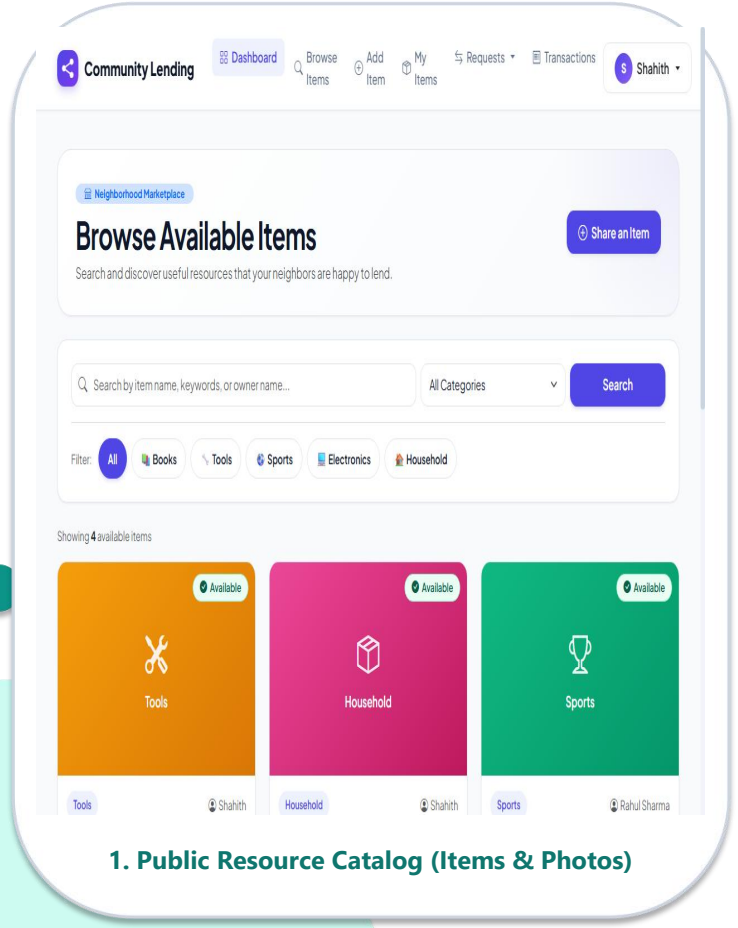
Step 3: Locks the item status to 'Lent' so nobody else can reserve it.

Step 4: Saves the new status as 'Approved' and informs the user.

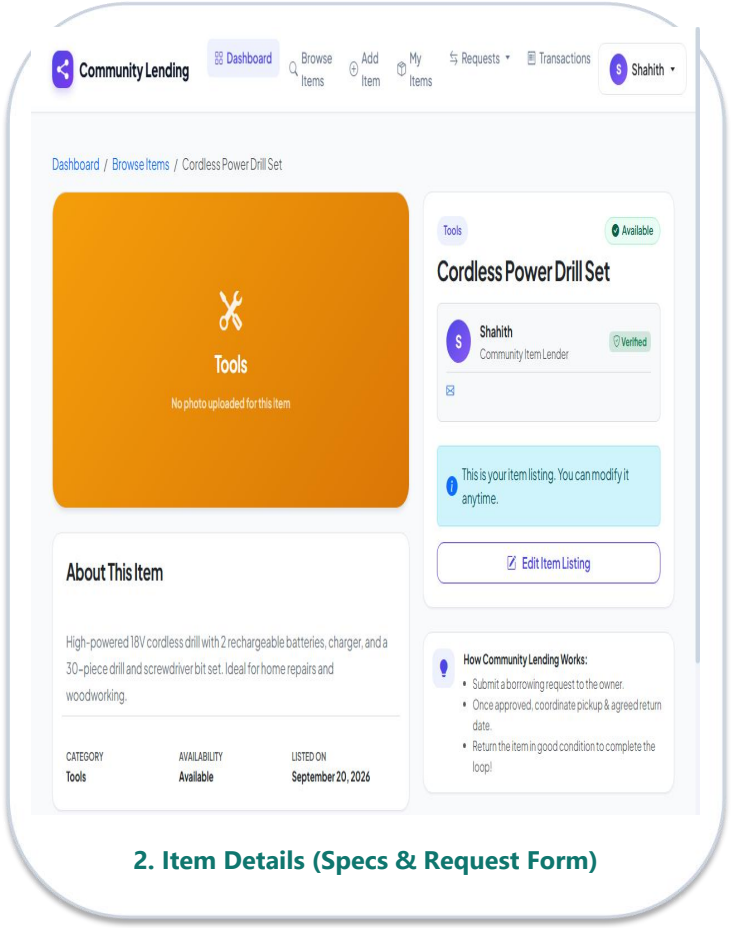
Complete Project Source Code:
https://github.com/Shahith2004/Community-Item-Lending-System

RESULT AND OUTPUT

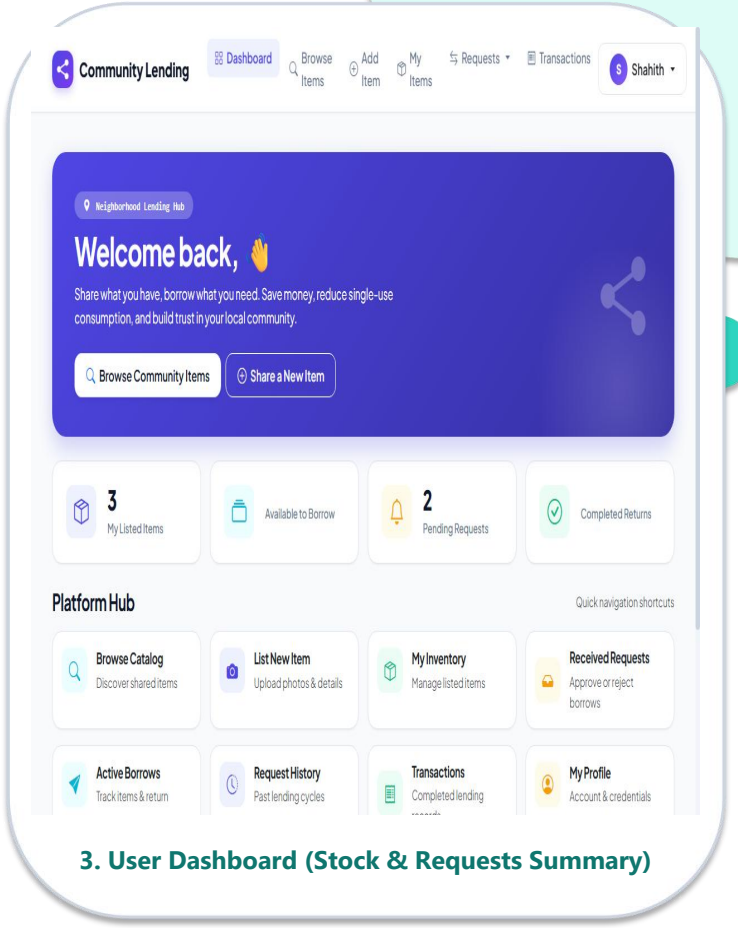
Actual Screenshots of the Running Website



1. Public Resource Catalog (Items & Photos)



2. Item Details (Specs & Request Form)



3. User Dashboard (Stock & Requests Summary)

Output: Clean, modern, and responsive website tested and running smoothly across all browsers.

CONCLUSION & FUTURE SCOPE

Summary of What Was Achieved & Next Steps

Conclusion — What Was Achieved

Working System: Successfully built a complete, easy-to-use website for community household item lending.

No Double Booking: Solved manual Excel and chat errors by locking item inventory automatically upon approval.

Paperless Tracking: Replaced slow group chats with automated online borrowing records and instant tracking.

Clutter & Cost Reduction: Connects power tools, lawnmowers, and ladders directly to neighbors, cutting household waste.

Secure & Cloud-Ready: Built on Python Flask and TiDB Cloud Serverless MySQL, ensuring robust data safety and 24/7 uptime.

Future Scope — Planned Enhancements

1. Mobile Application: Develop native Android and iOS mobile apps using Flutter so residents can easily snap photos and manage requests from their phones.

2. SMS & WhatsApp Alerts: Send quick text messages to resident phones when their borrow request gets approved, with automated return reminders.

3. Geolocation & Radius Search: Integrate interactive Google Maps or OpenStreetMap to filter items within a walkable neighborhood radius (e.g., within 2 km).

4. Community Trust Ratings: Introduce 5-star ratings and mutual feedback reviews to build greater confidence and reputation among neighborhood residents.

References — Books & Websites

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[3] Ramez Elmasri, Fundamentals of Database Systems (7th Edition), Pearson, 2016.

[4] Python Documentation https://docs.python.org/3/ (Python 3.10+ Reference Manual)

[5] Flask Web Framework https://flask.palletsprojects.com/ (Official Flask 3.0 Guide)

[6] MySQL Reference Manual https://dev.mysql.com/doc/ (MySQL 8.0 Protocol / TiDB Serverless)

[7] Bootstrap Documentation https://getbootstrap.com/ (Responsive UI Components Framework)

THANK YOU

Any Questions or Discussion?

Community Item Lending System (CILS)

ENROLMENT NUMBER : EA2532251010500

NAME OF STUDENT

MOHAMED SHAHITH AKEEL S A

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